Precision-fermented whey and egg proteins: what they cost, and where evidence would help

Memo · v2.1, audited and AI pre-reviewed (2026-10-01 22:02) · Pablo Antonio Moreno Casares · techno-economic model `pf-tea`, Humbird (2021) method, checked against NREL, Intelligen and Voutilainen designs · prepared with an AI assistant; not yet reviewed by human experts

Bottom line.

- **Today's public strains are far too expensive.** Food-grade β-lactoglobulin (BLG, whey) and ovalbumin (OVA, egg) cost $43 and $246 per kg protein at the best published titers. Whey protein traded at $9–$18 and dried egg white at $10–$21 per kg protein in 2015–2025 (2025 dollars).
- **At the titers companies' cost models assume, BLG is near whey.** At 42 g/L it costs $15.8 ($12.3 with the lighter accounting of NREL-style design reports); 3 of 21 annual whey export prices in 2015–2025 were above the first figure.
- **The gate is secretion, not titer.** $20/kg needs about 5–7 mg protein per g of cells per hour at the base cell density (the base design: 4.7). With today's biology and capital, design alone reaches $7.1/kg at 40 mg/g/h and $12.0 at 10. No published **K. phaffii** process is near either rate: compiled secretion rates stay below 1 mg/g/h and peak while cells grow fast, the best glucose fed-batch we found reached 2.1, and in the one near-zero-growth study, secretion fell as growth slowed.
- **Best case, not a floor.** With every assumed limit at its best, 400 m³ vessels and 100 kt/yr, BLG costs $4.1 and OVA $4.2 per kg, if cells secrete 40 mg/g/h; at 10 mg/g/h BLG costs $8.1, and at the best rate demonstrated in a public glucose fed-batch (2.1 mg/g/h) $24.5.
- **Lactoferrin shows the other side.** At a design titer of 10 g/L, PF lactoferrin costs $136–$178/kg, against $250–$1,585/kg reported for the dairy product. Intelligen's own model of the same route costs 2.8× ours to produce, mostly through its unit prices; even so it sits inside that range.

$/kg protein, 10 kt/yr, US	titer g/L	BLG	BLG light	OVA	OVA light
best public academic strain	13.9 / 2	$43	$32	$246	$176
base case	30	$21.2	$16.4	$21.3	$16.5
industry benchmark	42	$15.8	$12.3	$15.9	$12.4
physical-limit best case (100 kt/yr)	140 / 140	$4.1	$3.3	$4.2	$3.4

Capital. Fermenters are priced at an industrial vendor quote; the aseptic cell-culture vessels of our first version would add $9.8/kg to the base. The two accounting conventions differ by $4.7/kg, mostly because Humbird charges 9% of investment a year for maintenance, insurance and overheads and NREL-style reports under 2%.

Where evidence would help most, in our judgment:

1. **Secretion without growth.** How much BLG or OVA a strain secretes per g of cells per hour when it barely grows, over days, at 1–10 m³. It sets both today's cost and the best case.
2. **Open fed-batch data.** Titer, productivity and sugar yield measured **together**; the product yield in every cost estimate, ours included, is a design value.
3. **Real plant costs.** Capital, staffing and running costs of recent PF plants would settle the vessel price basis and the fixed-cost rule.
4. **The design ceiling.** Workable starting volumes, feed concentrations and O_2 transfer at scale.

EU. At Dutch sugar, energy, wage and construction prices the same plant costs about $3.1/kg more. A capital charge at 4% over 20 years instead of 7.5% over 10 would cut $3.9/kg; US-level sugar $2.2. These are cost effects on one plant, not policy recommendations.

Caveats. A production cost with a 14.6%/yr capital charge, not a market price. Excludes R&D, licensing, regulatory, sales, start-up, failed batches and first-of-a-kind premiums. Strain parameters rest on design values and company-reported averages (evidence grades B–C). The best public BLG titer uses an ethanol-fed production phase, priced here as glucose. Full report, audit log, simulated expert reviews and data: `pf-tea` repository.